

PRODUCT REQUIREMENTS DOCUMENT

AI-Enabled Detection of Exoplanets from Noisy Astronomical Light Curves

Version 1.0 | Hackathon Technical Specification | June 2026

Document Type PRD	Version 1.0	Status APPROVED	Priority P0 — Critical
~25k Light Curves Per TESS Sector	4 Signal Classes Transit / Eclipse / Blend / Other	≥90% Target Accuracy Classification F1 Score	3 Core Params Period · Depth · Duration

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1. Executive Summary

This Product Requirements Document defines the full technical specification for an AI-enabled end-to-end pipeline capable of autonomously detecting and classifying exoplanet transit signals within noisy astronomical light curves sourced from NASA's TESS mission archive.

Transit photometry—measuring minute stellar brightness dips as a planet crosses its host star—is one of the most productive methods of exoplanet discovery. However, real-world TESS light curves are polluted by instrumental noise, cosmic-ray events, stellar variability, and crowded-field blending, making automated, reliable detection a challenging machine-learning problem.

Core Value Proposition

Problem Space - Manually reviewing 20–30k light curves per sector is infeasible - Noise and contamination mask faint planetary signals - False positives (eclipsing binaries, blends) waste follow-up resources - Parameter estimation is time-consuming without automated tools	Solution Delivered - Automated ingestion and preprocessing of raw TESS data - ML classifier with ≥90% F1 across 4 signal categories - Transit parameter estimation (period, depth, duration) with uncertainties - Interactive dashboard with per-object confidence scoring
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2. Problem Statement & Scientific Background

2.1 Transit Photometry Fundamentals

Exoplanet transits manifest as periodic, box-shaped dips in a star's flux time series. The fractional depth $\Delta F/F \approx (R_p/R_\star)^2$ is typically 10^{-4} to 10^{-2} — far below typical photometric noise floors of 100–1000 ppm. Reliable detection therefore demands robust signal processing and statistical significance testing.

2.2 Signal Taxonomy

The pipeline must distinguish four phenomenologically distinct classes:

Class	Shape Feature	Physical Origin	Discriminating Trait
Transit	Flat-bottomed dip	Planet occults host star	Even depth; no secondary eclipse

Class	Shape Feature	Physical Origin	Discriminating Trait
Eclipsing Binary	V- or U-shaped dip	Stellar companion eclipse	Secondary eclipse ~50% depth
Blend (3rd-body)	Diluted dip	Background EB in aperture	Odd-even depth mismatch; centroid shift
Starspot / Variability	Sinusoidal / asymmetric	Stellar rotation / activity	No periodicity or non- periodic shape

2.3 Data Challenges

Key Technical Hurdles

- ▶ Photometric noise: TESS 2-minute cadence achieves ~100–10000 ppm scatter depending on stellar magnitude
- ▶ Systematic trends: spacecraft systematics, scattered light, momentum dumps corrupt portions of each sector
- ▶ Crowded fields: median TESS pixel subtends ~21 arcsec, blending multiple stars into one aperture
- ▶ Class imbalance: genuine transits account for <1% of all periodic signals in a typical sector
- ▶ Partial transits: edge-of-sector events yield incomplete light curves unsuitable for period estimation

3. Objectives & Success Metrics

3.1 Functional Objectives

The pipeline shall, without human intervention:

1. Download and preprocess high-cadence TESS light curves for a user-specified sector and camera/CCD combination
2. Identify all light curves exhibiting statistically significant periodic dip signatures
3. Classify each detected dip into: Planetary Transit, Eclipsing Binary, Blend, or Other Variability
4. Estimate orbital period, transit duration, and transit depth with associated uncertainties for transit candidates
5. Produce publication-quality visualizations and a ranked candidate catalog
6. Report signal-to-noise ratio and confidence level for every detection

3.2 Performance Targets

≥90% Precision Transit class	≥85% Recall Transit class	≥90% F1 Score Macro-avg all classes	<5% Period Error Median fractional
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4. System Architecture

The pipeline is structured as a six-stage sequential processor with feedback loops for quality control. Each stage is independently testable and can be swapped without affecting adjacent modules.

4.1 High-Level Architecture

Pipeline Flow

- ▶ Stage 1 — Data Acquisition: Bulk download of raw TESS FITS files from MAST archive for a given sector
- ▶ Stage 2 — Preprocessing & Detrending: Outlier rejection, gap filling, systematics removal (CBV / GP detrending)
- ▶ Stage 3 — Periodicity Detection: BLS (Box-Least Squares) periodogram to extract candidate dip signals
- ▶ Stage 4 — Feature Engineering: 35+ statistical, morphological, and spectral features per candidate
- ▶ Stage 5 — ML Classification: Ensemble classifier assigns class label and confidence score to each candidate
- ▶ Stage 6 — Parameter Estimation & Reporting: Transit model fitting, uncertainty quantification, catalog and plots

4.2 Technology Stack

Component	Technology	Description
Data Access	lightkurve 2.x / astroquery	MAST API integration; TESS FITS download and sector querying
Preprocessing	numpy / scipy / astropy	Time-series normalization, outlier sigma-clipping, gap interpolation
Detrending	wotan / celerite2	Cosine filtering, Gaussian Process systematics removal

Component	Technology	Description
Periodicity Search	transitleastsquares (TLS)	BLS/TLS periodogram with improved transit model sensitivity
Feature Extraction	tsfresh / custom	35+ statistical, morphological, and astrophysical features
ML Classification	scikit-learn / XGBoost / LightGBM	Gradient boosted ensemble; SMOTE oversampling for class imbalance
Deep Learning (opt.)	PyTorch / 1-D CNN	Raw flux ingestion; trained on ExoFOP labeled dataset
Parameter Fitting	batman-package / emcee	Mandel-Agol transit model; MCMC posterior sampling
Uncertainty Estimation	scipy.optimize + emcee	MCMC-derived credible intervals on all fitted parameters
Visualization	matplotlib / bokeh / plotly	Static publication figures and interactive HTML dashboard
Catalog Output	pandas / astropy Table	CSV + FITS catalog with all candidates and derived parameters
Reporting	Jinja2 / LaTeX/PDF	Automated 3-page report and per-target summary cards

5. Detailed Functional Requirements

5.1 Data Acquisition Module (DAM)

- SHALL query the MAST archive via astroquery.mast for all 2-minute cadence targets in a specified TESS sector/camera/CCD
- SHALL download SPOC-processed FITS files including SAP_FLUX, PDCSAP_FLUX, QUALITY flag columns
- SHALL filter on QUALITY bitmask to remove known bad cadences (bit flags 1, 2, 4, 8, 16, 512)
- SHALL support resumable downloads with local caching to avoid re-fetching already-downloaded sectors
- SHALL log download statistics: total targets, downloaded, failed, skipped

5.2 Preprocessing Module (PPM)

- SHALL normalize each light curve to unit median flux
- SHALL apply iterative sigma-clipping (3σ , 5 iterations) to remove outliers and cosmic rays
- SHALL fill small gaps (≤ 3 cadences) using linear interpolation; flag larger gaps
- SHALL apply wotan bi-weight or Gaussian Process detrending with tunable window length

- SHALL output normalized, detrended flux arrays with associated timestamps and uncertainty arrays

5.3 Periodicity Detection Module (PDM)

- SHALL run Transit Least Squares (TLS) on each detrended light curve
- SHALL search periods from 0.5 days to one-half the sector baseline
- SHALL return top-3 period candidates per light curve with power spectrum, SDE (Signal Detection Efficiency), and FAP
- SHALL apply a minimum SDE threshold of 7.0 to gate candidates into the classification stage
- SHALL compute primary and secondary eclipse depths for eclipsing binary discrimination

5.4 Feature Engineering Module (FEM)

For each BLS candidate, the system shall extract the following feature categories:

Statistical Features (15)

- BLS peak power / SDE
- False Alarm Probability (FAP)
- Transit depth, duration, impact parameter (b)
- Ingress/egress asymmetry ratio
- Odd-even depth ratio
- Out-of-transit scatter (σ_{OOT})
- Phase-folded light curve skewness & kurtosis

Morphological Features (20+)

- Secondary eclipse depth & phase offset
- Trapezoid fit residuals
- Centroid motion amplitude during transit
- Neighboring star flux contamination ratio
- Period aliasing indicators (P/2, 2P)
- Number of observed transits
- Galactic latitude and stellar type

5.5 Classification Module (CLM)

- SHALL train a gradient boosted ensemble (XGBoost + LightGBM) on the provided labeled dataset of known exoplanets, false positives, and eclipsing binaries
- SHALL apply SMOTE oversampling to address class imbalance during training
- SHALL output per-class probability scores [P_transit, P_EB, P_blend, P_other] summing to 1.0
- SHALL assign the highest-probability class as the predicted label
- SHALL achieve $\geq 90\%$ macro-averaged F1 score on held-out 20% test split
- SHALL optionally augment with a 1-D CNN trained on raw phase-folded flux arrays for ensemble stacking
- SHALL expose a threshold parameter to tune precision/recall trade-off per use case

5.6 Parameter Estimation Module (PEM)

- SHALL fit a Mandel-Agol transit model (batman-package) to phase-folded transit light curves
- SHALL estimate: Orbital period P [days], Transit depth $\delta = (R_p/R_\star)^2$, Transit duration T_{14} [hours], Time of first transit T_0 [BTJD], Impact parameter b , Limb darkening coefficients u_1, u_2
- SHALL run MCMC sampling (emcee, 1000+ steps, 32 walkers) for Bayesian posterior distributions
- SHALL report median parameter values and 16th/84th percentile credible intervals as uncertainties
- SHALL compute transit Signal-to-Noise Ratio: $SNR = \delta \times \sqrt{(N_{\text{transit}} \times T_{14}/\sigma)}$

5.7 Visualization Module (VIM)

Required Visual Outputs per Candidate

- ▶ Full raw light curve with detected transit epochs highlighted and quality-flagged cadences marked
- ▶ Detrended and normalized light curve showing preprocessing effect
- ▶ BLS/TLS periodogram with peak period annotation and FAP threshold line
- ▶ Phase-folded light curve with batman model overlay and 1σ uncertainty band
- ▶ Corner plot of MCMC posterior distributions for all fitted parameters
- ▶ Odd/even transit comparison plot for eclipsing binary discrimination
- ▶ Pixel-level centroid motion plot for blend detection

5.8 Output & Reporting Module (ORM)

- SHALL produce a ranked CSV catalog with columns: TIC ID, RA, Dec, TESS magnitude, predicted class, class probabilities, SNR, P , T_0 , δ , T_{14} , and all fitted uncertainties
- SHALL produce a FITS binary table version of the same catalog for compatibility with astronomical software
- SHALL generate a 3-page automated PDF summary report per the hackathon specification
- SHALL generate a per-target HTML summary page with all plots embedded for interactive review

6. AI Model Specification

6.1 Training Data Strategy

The model uses a two-source training approach to maximize generalization:

Source	Labels	Size (approx.)	Usage
ExoFOP-TESS Curated Set	Planet, EB, FP, BEB	~4,000 labeled	Primary training
NASA Exoplanet Archive	Confirmed planets	~5,500 targets	Positive augmentation
TESS SPOC TCE Catalog	TCE w/ dispositions	~10,000 entries	Feature pre-training
Synthetic injection-recovery	Generated transits	~2,000 injected	Minority class boost

6.2 Model Architecture

- Primary Model: XGBoost + LightGBM stacked ensemble trained on 35+ engineered features
- Secondary Model (optional): 1-D CNN processing 200-point phase-folded flux arrays (5 conv layers, global average pooling, 2 dense layers with dropout 0.3)
- Meta-learner: Logistic Regression combining both model probability outputs
- Calibration: Platt scaling applied to convert raw scores to calibrated probabilities

6.3 Training Protocol

- 80/10/10 split: training / validation / held-out test
- 5-fold cross-validation on training set for hyperparameter optimization (Optuna)
- SMOTE + Random Under-Sampling (RUS) to achieve approximately equal class representation
- Early stopping on validation F1 score with patience=20 rounds
- Model saved with versioning; automatic retraining trigger when new labeled data available

6.4 Uncertainty Quantification

Multi-Layer Uncertainty Approach

- ▶ Aleatoric uncertainty: Monte Carlo Dropout (T=50 forward passes) captures data-inherent noise
- ▶ Epistemic uncertainty: Model confidence from calibrated probability spread across classes
- ▶ Parameter uncertainty: MCMC credible intervals from emcee posterior sampling (P, δ , T14, T0)
- ▶ Detection significance: SDE and analytical False Alarm Probability from TLS periodogram
- ▶ Ensemble disagreement: Variance between XGBoost and LightGBM outputs as reliability indicator

7. Enhanced Features for Improved Feasibility

The following features go beyond the baseline specification to significantly improve pipeline robustness, accuracy, and usability:

7.1 Automated Quality Gating

- Stellar type filter: prioritize FGK dwarf stars ($\log g > 4.0$) where transits are detectable
- Sector coverage filter: skip targets with < 13.5 days of usable data
- Noise floor pre-check: skip targets with $\sigma_{\text{PDCSAP}} > 5 \times$ expected photon noise (likely contaminated)
- Crowding metric: use TESS contamination ratio from TIC catalog to flag blended apertures

7.2 Injection-Recovery Testing

- Automated injection of synthetic transit signals (batman-generated) into real light curves
- Recovery rate curves as a function of period, depth, and stellar magnitude
- Provides pipeline completeness estimates essential for occurrence rate studies
- Enables model calibration and bias correction at the population level

7.3 Multi-Sector Support

- Automatic stitching of light curves across consecutive TESS sectors for improved phase coverage
- Period uncertainty improves as $\sqrt{N_{\text{sectors}}}$ with multi-sector data
- Enables detection of long-period planets (> 27 days) requiring sector baseline exceeding one sector

7.4 Centroid Analysis

- Pixel-level centroid motion analysis during transit events to flag background eclipsing binaries
- PRF (Pixel Response Function) fitting to quantify flux contamination from neighboring stars within 1 arcminute
- Automated flag if centroid shifts by more than 0.5 arcsec during transit

7.5 Stellar Property Integration

- Automatic cross-match with TIC (TESS Input Catalog) and Gaia DR3 for stellar parameters
- Physical radius estimation: R_p/R_{Earth} derived from transit depth + stellar radius

- Habitable zone proximity score computed for each transit candidate
- Spectral type classification enables planet-to-host mass ratio estimation

7.6 Active Learning Loop

- Flag low-confidence candidates (max class probability < 0.65) for human review
- Provide a lightweight labeling web interface (Streamlit) for expert vetting
- Labeled outputs automatically appended to training set and trigger model fine-tuning
- Tracks annotation history and inter-annotator agreement for quality control

7.7 Batch Processing & Scalability

- Parallelized preprocessing and feature extraction using Python multiprocessing (joblib)
- Estimated throughput: ~500 light curves / minute on 8-core CPU
- Full sector processing (25k targets) expected in under 60 minutes on standard hardware
- GPU acceleration optionally available for CNN forward passes (PyTorch DataLoader)

8. Implementation Roadmap

Phase	Name	Key Deliverables	Duration	Priority
Phase 0	Setup & Data Ingestion	Env setup, MAST download, EDA on 1 sector	Day 1	P0
Phase 1	Preprocessing & BLS	Detrending, TLS search, candidate list	Days 1–2	P0
Phase 2	Feature Engineering	35+ features extracted per candidate	Day 2	P0
Phase 3	ML Classifier	Trained ensemble, ≥90% F1, SMOTE	Days 2–3	P0
Phase 4	Parameter Fitting	batman + emcee MCMC, credible intervals	Day 3	P0
Phase 5	Visualization & Report	Plots, catalog CSV, 3-page PDF report	Day 3–4	P0
Phase 6 (Bonus)	Enhanced Features	Centroid analysis, injection-recovery, active learning	Day 4	P1

9. Testing & Validation Strategy

9.1 Unit Tests

- Each pipeline module has independent unit tests with mock FITS data
- BLS module validated against TOI (TESS Object of Interest) catalog known periods
- Feature extraction tested for determinism and numerical stability

9.2 Integration Tests

- End-to-end pipeline run on TESS Sector 1, Camera 1, CCD 1 (reference sector with many confirmed planets)
- Benchmark against published TOI catalog: require $\geq 80\%$ of confirmed planets to be recovered
- False positive rate on background stars (non-variable) must be $< 2\%$

9.3 Model Validation

- Confusion matrix on held-out test set reported for all four classes
- Precision-Recall curves for transit class at multiple thresholds
- SHAP feature importance analysis to ensure physically meaningful features drive predictions
- Calibration plot (reliability diagram) to verify probability estimates are well-calibrated

10. Risks & Mitigations

Risk	Impact	Severity	Mitigation
MAST download bandwidth limits	Sector download incomplete	High	Pre-cache data; use MAST bulk download tools
Severe class imbalance in labels	Classifier biased to majority class	High	SMOTE + class-weighted loss; injection-recovery augmentation
Poor detrending masking real transits	Missed detections	Medium	Compare multiple detrending algorithms; inject-recover test
MCMC non-convergence	Unreliable parameter uncertainties	Medium	Set max iteration limits; fall back to least-squares fit
Unrecognized signal morphologies	Misclassification	Low	Out-of-distribution detector; flag uncertain cases for human review

11. Dependencies & Environment

11.1 Python Package Requirements

All dependencies are open-source and pip-installable. Recommended Python version: 3.10+.

Package	Version	Purpose
lightcurve	≥2.4	TESS FITS access, light curve utilities
transitleastsquares	≥1.0	TLS periodogram for transit detection
batman-package	≥2.4	Mandel-Agol transit model generation
emcee	≥3.1	MCMC sampler for Bayesian parameter estimation
wotan	≥1.10	Light curve detrending algorithms
celerite2	≥0.2	Gaussian Process systematics correction
xgboost / lightgbm	≥1.7 / ≥3.3	Gradient boosted classification ensemble
scikit-learn	≥1.3	ML utilities, SMOTE, calibration
imbalanced-learn	≥0.11	SMOTE oversampling for class balance
astropy / astroquery	≥5.3 / ≥0.4	FITS I/O, coordinate tools, MAST API
tsfresh	≥0.20	Automated time-series feature extraction
bokeh / plotly	≥3.3 / ≥5.17	Interactive dashboard visualization
torch (optional)	≥2.0	1-D CNN for raw flux classification
optuna	≥3.4	Hyperparameter optimization for ML models
shap	≥0.44	Model explainability / feature importance

12. Acceptance Criteria

The system is considered complete and submission-ready when ALL of the following criteria are met:

Go / No-Go Checklist

- ▶ ☐ Pipeline processes a full TESS sector (25k+ targets) end-to-end without crashing
- ▶ ☐ BLS/TLS detection recovers ≥80% of known TOIs present in the processed sector at SDE ≥ 7
- ▶ ☐ Classifier achieves ≥90% macro-F1 on held-out test set across all 4 classes
- ▶ ☐ Period estimates for confirmed planets within 5% of catalog values
- ▶ ☐ Transit depth and duration estimates within 10% of published values
- ▶ ☐ Every candidate receives a calibrated confidence score between 0 and 1

- ▶ [] All 7 required visualizations generated per transit candidate
- ▶ [] Output CSV catalog includes all required columns with no null values for transit candidates
- ▶ [] 3-page PDF report generated automatically by the pipeline
- ▶ [] Runtime for a full sector ≤ 90 minutes on 8-core CPU without GPU

13. Appendix — Directory Structure

Recommended project layout for reproducibility and modularity:

Path	Contents
<code>exoplanet_pipeline/</code>	Project root
├─ <code>data/raw/</code>	Downloaded TESS FITS files
├─ <code>data/processed/</code>	Detrended light curves (numpy .npz)
├─ <code>data/catalogs/</code>	TOI, TIC, ExoFOP reference catalogs
├─ <code>models/</code>	Trained model artifacts (.pkl, .pt)
├─ <code>src/acquisition.py</code>	MAST download & FITS parsing
├─ <code>src/preprocessing.py</code>	Detrending, normalization, outlier removal
├─ <code>src/detection.py</code>	TLS/BLS periodogram wrapper
├─ <code>src/features.py</code>	Feature extraction for ML
├─ <code>src/classifier.py</code>	Train, predict, evaluate ensemble model
├─ <code>src/fitting.py</code>	batman + emcee parameter estimation
├─ <code>src/visualization.py</code>	All plotting functions
├─ <code>src/report.py</code>	PDF report generation
├─ <code>notebooks/</code>	EDA and result exploration notebooks
├─ <code>tests/</code>	Unit and integration tests
├─ <code>outputs/</code>	Candidate catalogs, plots, reports
└─ <code>run_pipeline.py</code>	CLI entry point for full pipeline

Document prepared for Hackathon Submission | All specifications subject to team capacity constraints